

BARKHAN, Pakistan

WMO: 416850

Lat: 29.883N

Lon: 69.717E

Elev: 3602

StdP: 12.88

Time Zone: 5.00 (E05)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 36.6	(c) 38.8	(d) 4.3	(e) 7.8	(f) 55.5	(g) 9.4	(h) 10.1	(i) 55.9	(j) 6.1	(k) 53.3	(l) 5.2	(m) 54.2	(n) 0.9	(o) N/A	(p) 0.325

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 21.0	(c) 102.2	(d) 67.9	(e) 99.9	(f) 68.3	(g) 97.8	(h) 68.4	(i) 77.3	(j) 91.7	(k) 76.3	(l) 90.6	(m) 75.4	(n) 89.6	(o) 2.3	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.6	143.4	85.9	72.6	138.4	85.0	71.6	133.6	84.3	43.9	92.1	42.8	90.2	41.8	89.6	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.5	(b) 5.5	(c) 4.4	(e) 34.0	(f) 106.2	(g) 3.1	(h) 2.2	(i) 31.8	(j) 107.8	(k) 30.0	(l) 109.1	(m) 28.2	(n) 110.3	(o) 25.9	(p) 111.9		
(a) 7.5	(b) 5.5	(c) 4.4	(e) 28.2	(f) 79.9	(g) 1.9	(h) 1.6	(i) 26.8	(j) 81.0	(k) 25.7	(l) 82.0	(m) 24.6	(n) 82.8	(o) 23.2	(p) 84.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	71.5	50.7	54.8	64.2	74.7	84.2	87.2	85.2	83.4	81.1	74.2	63.2	54.2
	DBStd	13.79	4.38	6.35	7.28	6.47	5.55	4.59	3.71	3.14	3.52	4.61	4.60	4.74
	HDD50	78	45	19	1	0	0	0	0	0	0	0	0	13
	HDD65	1264	442	289	104	6	0	0	0	0	0	1	87	335
	CDD50	7932	67	153	441	742	1061	1116	1093	1037	932	750	397	143
	CDD65	3646	0	4	79	298	596	666	628	572	482	286	34	1
	CDH74	44426	1	49	754	3153	8066	9299	8168	6623	5280	2690	317	26
	CDH80	22937	0	5	221	1408	4639	5536	4372	3140	2547	1037	30	2
Wind	WSAvg	1.1	0.9	1.3	1.1	1.4	1.5	1.3	1.0	1.1	0.9	0.8	1.1	0.9
Precipitation	PrecAvg	16.4	0.5	0.9	1.2	1.4	1.1	2.0	4.1	3.0	1.4	0.5	0.2	0.3
	PrecMax	30.0	2.0	3.2	4.2	5.1	5.7	6.5	8.7	6.3	5.2	5.5	1.0	2.0
	PrecMin	4.0	0.0	0.0	0.0	0.0	0.0	0.1	0.6	0.2	0.0	0.0	0.0	0.0
	PrecStd	5.8	0.6	0.9	0.9	1.3	1.2	1.6	2.0	1.7	1.3	1.1	0.3	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.7	79.3	89.8	98.0	104.0	106.1	101.8	98.5	99.0	94.1	83.4	77.4
		MCWB	48.8	53.7	59.7	63.5	66.4	69.4	73.2	74.6	68.1	65.6	56.9	53.4
	2%	DB	67.6	74.9	85.8	94.4	101.3	103.4	99.3	96.1	95.5	91.3	80.1	72.6
		MCWB	48.0	52.9	59.2	63.2	64.7	69.0	73.3	73.8	68.5	65.2	57.0	51.1
	5%	DB	64.7	71.4	81.8	91.3	99.0	101.0	96.8	94.2	93.4	88.8	77.8	68.7
		MCWB	46.3	52.4	57.7	61.9	64.3	68.8	73.8	73.6	68.7	63.5	55.8	49.0
	10%	DB	61.9	67.5	77.8	88.0	96.4	98.3	94.4	92.1	91.2	85.9	75.3	65.6
		MCWB	45.5	50.3	55.7	61.1	63.5	68.3	73.6	73.7	69.0	61.7	54.6	47.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.8	58.2	63.6	69.7	71.4	78.1	78.5	79.1	77.2	70.8	62.4	55.8
		MCDB	67.7	71.7	83.9	89.1	95.5	96.6	93.5	93.1	88.6	85.2	73.9	74.4
	2%	WB	49.7	56.3	61.0	66.4	69.3	75.9	77.1	77.5	74.9	68.7	60.4	52.7
		MCDB	62.9	69.8	81.3	87.2	94.3	94.2	91.8	90.9	86.5	85.7	75.0	68.3
	5%	WB	48.1	54.2	58.7	64.3	67.8	74.4	76.2	76.3	73.3	66.8	58.4	50.5
		MCDB	60.9	68.6	76.8	85.9	91.6	92.5	90.9	89.4	85.0	83.8	74.0	65.6
	10%	WB	46.6	51.6	57.1	62.6	66.2	73.0	75.4	75.3	71.8	64.9	56.4	48.6
		MCDB	59.4	65.7	75.4	83.7	90.5	90.8	90.2	88.0	84.9	81.3	72.1	63.5
Mean Daily Temperature Range	5% DB	MDBR	20.8	20.4	20.5	20.8	21.5	21.0	18.7	17.7	20.4	22.4	21.6	21.5
		MCDBR	24.5	24.2	24.1	24.0	23.4	23.2	20.8	20.0	22.5	23.9	24.2	25.3
		MCWBR	11.5	11.0	9.4	9.2	8.1	8.1	6.6	6.6	8.0	9.1	11.1	12.8
	5% WB	MCDBR	22.0	21.4	22.0	22.2	22.8	22.0	19.4	18.5	19.4	22.8	22.2	23.3
		MCWBR	11.9	10.8	10.6	10.5	9.5	9.0	7.8	7.7	8.2	9.3	11.3	12.6
Clear-Sky Solar Irradiance	taub	0.375	0.420	0.447	0.502	0.524	0.633	0.719	0.639	0.506	0.454	0.432	0.375	
	taud	2.111	2.011	1.956	1.855	1.816	1.632	1.520	1.663	1.893	1.926	1.946	2.112	
	Ebn at Noon	265	263	265	255	249	222	203	218	245	248	241	258	
	Edh at Noon	42	50	57	65	69	82	91	78	60	54	49	40	
All-Sky Solar Radiation	RadAvg	1158	1394	1797	2108	2301	2206	2009	1932	1870	1637	1276	1110	
	RadStd	66	122	105	87	67	76	94	117	70	56	67	48	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page